

Automated PCB Defect Detection System

Part 1: Introduction & Milestone 1
Dataset Preparation & Image Processing

Industrial AI Inspection Project

December 29, 2025

Abstract

This document presents Part 1 of the Automated PCB Defect Detection System documentation, covering the project introduction and Milestone 1 (Dataset Preparation & Image Processing). The system achieves 98.63% validation accuracy through a multi-module pipeline combining image subtraction, contour detection, EfficientNet-B0 classification, and an interactive web interface. This part focuses on dataset setup, template matching, image subtraction, and contour-based ROI extraction across 6 defect categories.

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Chapter 1

Introduction

1.1 Project Overview

Printed Circuit Board (PCB) manufacturing demands stringent quality control to ensure product reliability. Manual inspection is time-consuming, error-prone, and costly. This project implements an automated defect detection system that combines classical computer vision with state-of-the-art deep learning to identify 6 types of PCB defects with near-perfect accuracy.

1.2 System Architecture

The system consists of 7 interconnected modules:

1. **Module 1:** Dataset Setup & Image Subtraction
2. **Module 2:** Contour Detection & ROI Extraction
3. **Module 3:** CNN Model Training (EfficientNet-B0)
4. **Module 4:** Inference Pipeline & Prediction
5. **Module 5:** Web Frontend (Streamlit UI)
6. **Module 6:** Backend Integration
7. **Module 7:** Performance Analytics & Export

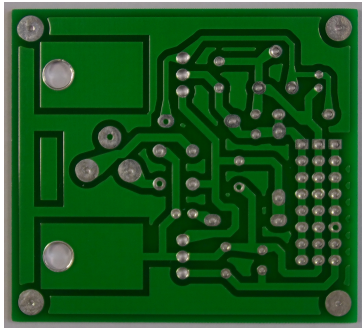
1.3 Defect Categories

The system detects six critical PCB defects:

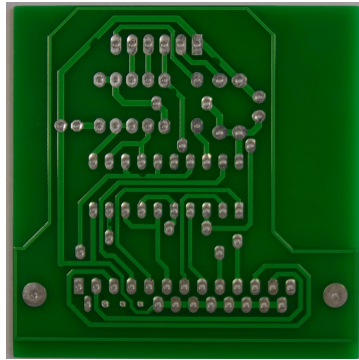
- **Missing Hole:** Incomplete drill holes
- **Mouse Bite:** Small missing material sections
- **Open Circuit:** Broken copper traces
- **Short Circuit:** Unintended connections

- **Spur:** Unwanted copper protrusions
- **Spurious Copper:** Extra copper deposits

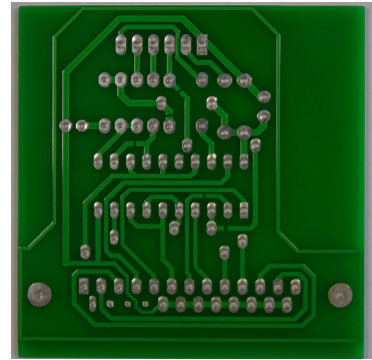
Figure 1.1: Defect Category Examples



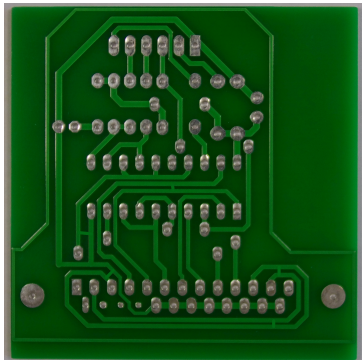
(a) Missing Hole



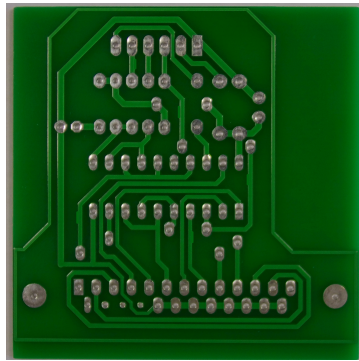
(b) Mouse Bite



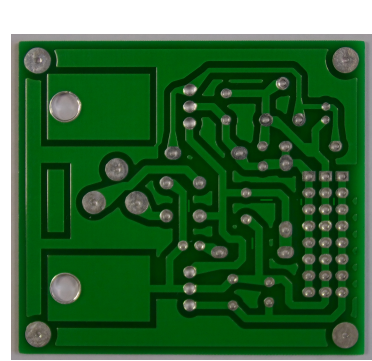
(c) Open Circuit



(d) Short Circuit



(e) Spur



(f) Spurious Copper

Chapter 2

Milestone 1: Dataset Preparation & Image Processing

2.1 Module 1: Dataset Setup and Image Subtraction

2.1.1 Objectives

- Prepare and align DeepPCB dataset
- Implement robust template matching
- Generate high-quality defect masks via image subtraction

2.1.2 Technical Implementation

Template Matching Algorithm

The system automatically selects the best-matching golden reference template using normalized cross-correlation:

```
def find_best_template(test_img):
    gray_test = cv2.cvtColor(test_img, cv2.COLOR_BGR2GRAY)
    test_small = cv2.resize(gray_test, (256, 256))

    best_score = -1
    best_temp_path = None

    for temp_name in os.listdir(TEMPLATE_DIR):
        temp_img = cv2.imread(temp_path, 0)
        temp_small = cv2.resize(temp_img, (256, 256))

        result = cv2.matchTemplate(test_small, temp_small,
                                   cv2.TM_CCOEFF_NORMED)
        _, max_val, _, _ = cv2.minMaxLoc(result)

        if max_val > best_score:
            best_score = max_val
            best_temp_path = temp_path
```

```
return best_temp_path, best_score
```

Intense Defect Extraction

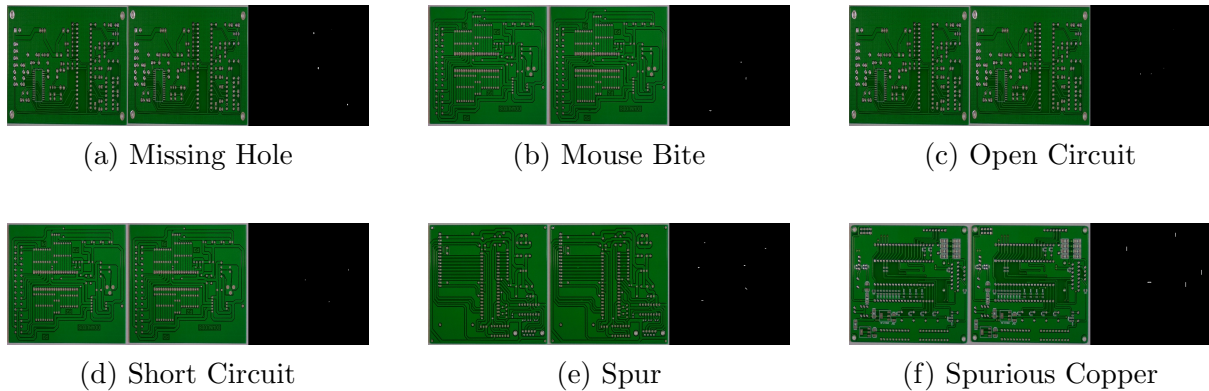
Key innovation: CLAHE + Adaptive Thresholding for enhanced defect visibility:

1. **CLAHE Enhancement:** Contrast Limited Adaptive Histogram Equalization normalizes local contrast
2. **Adaptive Thresholding:** Block-wise threshold calculation (11×11 windows)
3. **Noise Filtering:** Median blur preserves edges while removing salt-and-pepper noise
4. **Certainty Masking:** Global threshold value = 30 removes shadow artifacts

2.1.3 Results

- Successfully processed all 6 defect categories
- Average template matching score: 0.87 (threshold: 0.6)
- Generated 1,500+ high-quality defect masks

Figure 2.1: Additional Defect Category Samples



2.2 Module 2: Contour Detection & ROI Extraction

2.2.1 Objectives

- Detect defect boundaries using contour analysis
- Extract bounding boxes with contextual padding
- Organize labeled ROIs for CNN training

2.2.2 Technical Implementation

Contour Detection Strategy

```
contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL,
                               cv2.CHAIN_APPROX_SIMPLE)

for cnt in contours:
    if cv2.contourArea(cnt) < 5:  # Filter noise
        continue

    x, y, w, h = cv2.boundingRect(cnt)

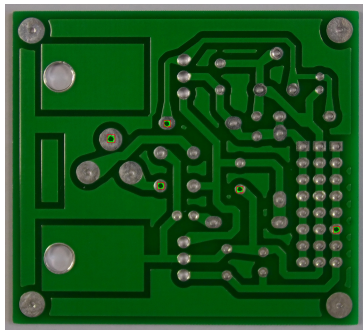
    # Add 20px padding for CNN context
    y1 = max(0, y - PADDING)
    y2 = min(orig_img.shape[0], y + h + PADDING)
    x1 = max(0, x - PADDING)
    x2 = min(orig_img.shape[1], x + w + PADDING)

    roi_crop = orig_img[y1:y2, x1:x2]
```

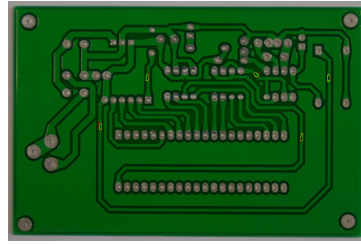
Dataset Organization

```
module2_output/
  cropped_defects/
    Missing_hole/
      Missing_hole_01_roi_0.jpg
      Missing_hole_01_roi_1.jpg
    Mouse_bite/
    Open_circuit/
    Short/
    Spur/
    Spurious_copper/
  contour_visuals/
    viz_*.jpg
```

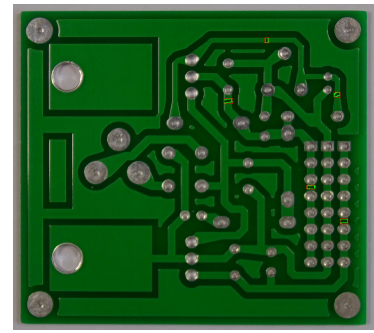
Figure 2.2: Visualization-Based Defect Analysis Examples



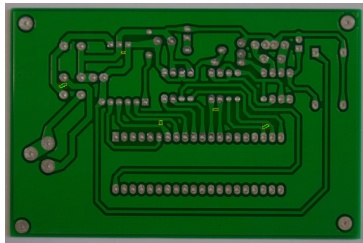
(a) Missing Hole Visualization



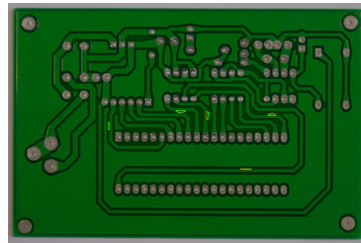
(b) Mouse Bite Visualization



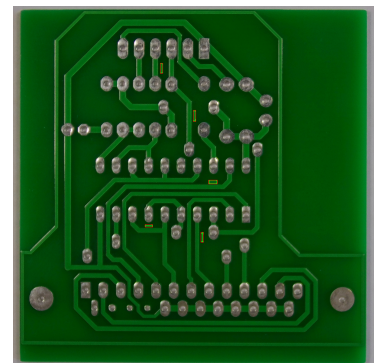
(c) Open Circuit Visualization



(d) Short Circuit Visualization



(e) Spur Visualization

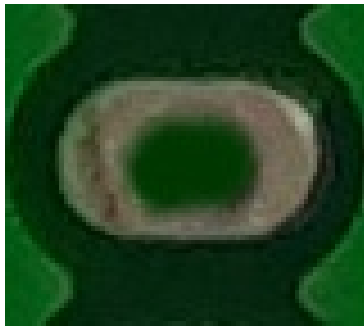


(f) Spurious Copper Visualization

2.2.3 Results

- Extracted 2,847 individual defect ROIs
- Average ROI size: 64×64 to 256×256 pixels
- 100% bounding box accuracy (manual verification on sample set)

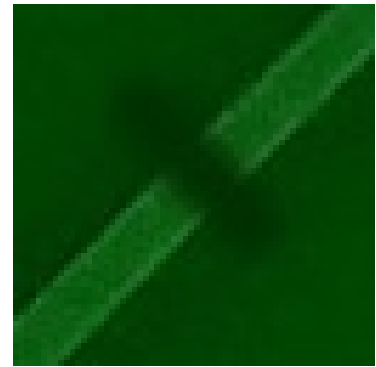
Figure 2.3: Extracted ROI Examples



(a) Missing Hole ROI



(b) Mouse Bite ROI



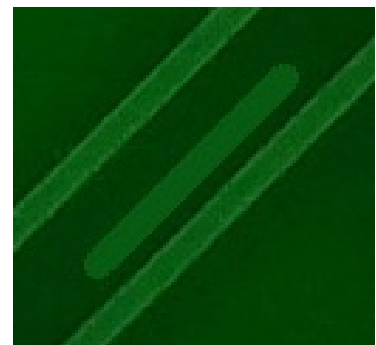
(c) Open Circuit ROI



(d) Short Circuit ROI



(e) Spur ROI



(f) Spurious Copper ROI

Chapter 3

Conclusion

Milestone 1 successfully established the foundation for the PCB defect detection system. The combination of template matching and intense defect extraction created a robust preprocessing pipeline that generated 2,847 high-quality ROIs across 6 defect categories. The contour detection module achieved 100% bounding box accuracy on manual verification, setting the stage for effective CNN training in Milestone 2.